

Does the Selected Object Reach the Reader? Auditing Identity Handoffs in Grounded Language-Model Pipelines

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Abstract

Grounded language-model pipelines can be divided into three stages: selecting an object, retrieving passages for it, and using that evidence to answer. If the selected object must reach the reader, losing it breaks the handoff. Benchmark recall checks the dataset-linked object, which can differ. We audit 600 HybridQA questions across three selector families. On 1,463 resolvable records where the selected object matches the dataset-traced passage, exact key lookup and exact title matching return the object every time. With every ranked rule given the same decoded selected title, body-only BM25 omits it on 389 records (26.6%) at cutoff five, while hybrid retrieval with reranking omits it on 14 (1.0%). The two identities differ on 329 of 1,792 resolvable records. With original-question rankings, their top-five checks disagree on 106 records (5.9%). Frozen reader comparisons associate the aligned object’s presence with 28.6 to 31.0 points higher exact match. In a deliberately selected 64-item cohort, removing that passage sharply lowers exact match, while removing a similar-length comparison passage does not reproduce the drop. We release the Returned-Object Profile (ROP), an executable record of the target, returned-ID field, cutoff, membership rule, and complete expected population, with data and an offline replay.

1 Introduction

Some grounded language-model pipelines turn a structured choice into text evidence. An entity linker can select a page, and a table reasoner can choose a linked record. Retrieval then turns that choice into passages for a reader (Li et al., 2020; Chen et al., 2020; Shavarani and Sarkar, 2025). The reader receives only the returned passages. If

the selected object drops from that list, the rest of the pipeline can no longer use it.

A pipeline can fail because selection is poor, because a required identity handoff breaks, or because the reader fails to use returned evidence. These causes call for different fixes. A single top- k recall score cannot separate them unless it names both the object being checked and the component responsible for returning it.

We call the selected object U and the passage linked by the dataset G . Dataset-trace recall checks benchmark coverage with G . Selected-object return checks whether a selector-to-retriever handoff delivers U . Semantic correctness remains its own measurement. When $U = G$, both checks ask about the same object. When the identities differ, the score must say whether it describes benchmark coverage or delivery of the recorded selection.

Our main question is what happens after a pipeline already holds a selected object ID. It can look up that ID directly or use its title or text to retrieve a broader passage set. These handoff rules preserve the recorded identity at different rates. Figure 1 shows the interface and the main result.

We evaluate this handoff on 600 HybridQA questions. Three model families each process all 600 questions, giving 1,800 planned model-question records. On 1,463 records, the selected object agrees with HybridQA’s linked passage and resolves in the frozen corpus. Exact key dereference and exact decoded-title equality return all 1,463 objects. At cutoff five, body-only BM25 omits 389, ColBERTv2 omits 20, and the final hybrid reranker omits 14. With the selected ID held fixed, the handoff rule determines whether that ID appears in the evidence list.

The 329 resolvable records where U and G differ show why this boundary needs its own target. They come from 160 questions. Original-question BM25 receives only the question, yet the two targets give different top-five labels on 106 of 1,792 records.

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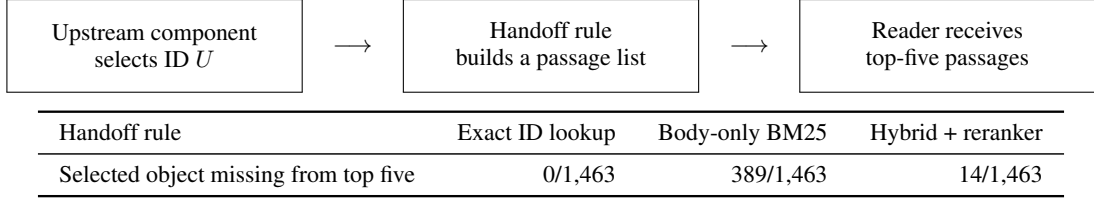


Figure 1: For each of 1,463 aligned, resolvable records, the selected object is held constant across handoff rules. The rule determines whether that object appears in the reader’s top five passages. Both ranked rules receive the same decoded selected title.

Queries that name the selected object produce more disagreements, showing why query construction belongs in the evaluation record. This audit measures which identity reaches the reader. Semantic quality remains a separate measurement.

We also follow aligned objects into the reader. In recorded five-passage comparisons, exact match is 28.6 to 31.0 points higher when the shared object is present. The contexts differ in other ways, so this result is associative. We then make fresh calls for 64 high-contrast cases chosen before the new outcomes. Exact match falls when the aligned passage is removed. Removing a similar-length nontraced passage does not reproduce the loss.

We make three contributions. First, we compare nine ways to turn a selected ID into reader evidence and measure whether the object reaches the top five. Second, we score selected-object return and dataset-trace recall against the same frozen rankings across four query forms, five retrieval stacks, and three cutoffs. Third, we release the ROP schema and tested reference evaluator, which encode the target, expected population, identity resolver, output field, cutoff, membership rule, and run result. The released artifact reproduces every reported aggregate offline from the frozen rankings and reader records. The project website is available at <https://opensciagent.github.io/rop>.

2 Measuring an Identity Handoff

Some grounding pipelines carry a selected object ID into retrieval. The upstream component emits that ID, and retrieval returns a ranked list of passage or chunk IDs. This gives the boundary a concrete input and output. We map both through one identity resolver. Selected-object return at cutoff k succeeds when one of the first k returned IDs maps to U or to a declared acceptable alternative. Dataset-trace recall uses the same ranking and membership rule with G as the target.

The Returned-Object Profile (ROP) records this

evaluation. It begins with the interface claim. When a documented policy requires the selected object to remain available after retrieval, ROP turns that rule into a complete-run check. The same fields can describe an interface with no selected-return requirement. In either mode, ROP isolates one boundary: whether the declared target appears in the returned IDs.

For a developer who requires the selected object in the reader context, these counts distinguish identity-preserving lookup from ranked evidence construction that needs an explicit return check.

Table 1: The five decisions recorded by a ROP declaration. Appendix A gives the complete fields and verdict order.

Evaluation question	Decision recorded in the declaration
Which claim applies?	The documented interface and whether it requires selected-object return.
Which object counts?	The selected target or its declared acceptable alternatives.
Which records belong?	The complete list of expected record IDs.
What counts as returned?	The identity map, returned-ID field, cutoff, and any or all membership rule.
How is the result decided?	One verdict for every expected record and one status for the complete run.

A resolver maps each raw key, alias, or chunk ID to one shared object identity. The returned-ID field and cutoff identify the part of the output being checked. The any rule accepts one returned ID that resolves to an acceptable target. The all rule requires every raw resolver member for one acceptable target. Unknown IDs do not match, and duplicate IDs still consume ranked positions.

The expected population is equally important. Suppose a run expects two rows and both pass. If one row disappears, the pass fraction over the surviving row is still 1.0. The run is only half complete.

Table 2: Population accounting in the synthetic hard-mode fixture. The observed-row pass fraction is identical, while the run status detects the missing expected record.

Case	Expected	Observed	Pass	Pass fraction	Run status
Complete	2	2	2	1.0	RUN_PASS
One missing	2	1	1	1.0	RUN_UNVERIFIABLE

Table 3: How the reference evaluator accounts for common observations. Record accounting and run status remain separate so that an omitted target, unusable output, and incomplete run cannot collapse into one score.

Observed condition	Record accounting	Run status
Requirement inactive	NOT_APPLICABLE	NOT_APPLICABLE
Target returned in a complete run	PASS	RUN_PASS
Target absent	FAIL	RUN_FAIL
Returned IDs missing or malformed	UNVERIFIABLE	RUN_UNVERIFIABLE
Expected record missing	Missing population entry	RUN_UNVERIFIABLE
Duplicate or unexpected record ID	Invalid population	RUN_INVALID

ROP lists every expected record ID, assigns one verdict to each expected record, and marks missing, duplicate, malformed, or unexpected observations before it reports the run result.

Table 2 shows the missing-row case. The reference suite also separates a failed membership check from an audit that cannot be scored. A resolvable target outside the cutoff receives FAIL. A missing or malformed returned-ID list receives UNVERIFIABLE. Missing expected records make the run unverifiable, while duplicate or unexpected record IDs invalidate it. These outcomes point to different parts of the pipeline.

A membership failure means that a usable returned list omits the target. An unverifiable record lacks a usable returned list. A missing expected ID makes the run incomplete, while duplicate or unexpected IDs make its observed population inconsistent with the declaration. Keeping these outcomes separate tells a developer whether to inspect retrieval, logging, or run assembly.

The released synthetic fixtures exercise all seven record verdicts and five run statuses. They establish the behavior of the reference evaluator. The next section reports a retrospective audit over 1,800 frozen HybridQA records using the same target, resolver, cutoff, and population concepts. The artifact replays that audit offline from hash-bound rankings and reader records. Appendix A gives the schema

Table 4: Complete accounting for the 1,800 planned model-question records. The four outcome columns partition every row. The $U = G$ and $U \neq G$ columns contain corpus-resolvable selected IDs.

Model	Planned	Invalid output	ID absent from corpus		$U = G$	$U \neq G$
Claude	600	0		1	456	143
GPT	600	6		0	509	85
Gemini	600	0		1	498	101
Total	1,800	6		2	1,463	329

and verdict order. Appendix B gives the execution settings and replay steps.

3 HybridQA Identity-Handoff Audit

3.1 Study design

HybridQA supports an identity-handoff audit because its table cells link to Wikipedia passages (Chen et al., 2020). We retain questions with one answer-containing linked passage, no answer in the table, no linked-cell label in the question, and a selection rule fixed before model outputs are inspected. These filters retain 7,201 training questions. A salted hash selects 600, with at most one question per table. The corpus contains 286,270 passages linked from 15,314 tables.

Claude Opus 4.7, GPT-5.5, and Gemini 3.1 Pro Preview each receive a question and its full table. Each model is asked to return a selected `/wiki/...` key U and a free-form retrieval fact F . The selected key is the upstream identity in the frozen pipeline. Each query condition determines whether retrieval receives that identity. HybridQA’s annotated link G remains a separate trace object. Each model-question pair contributes one recorded selection. The three model families describe three frozen pipelines, so this design does not measure how a selector varies across repeated samples. Six of the 1,800 selector rows are invalid, and two valid keys do not resolve in the corpus. All eight remain in the accounting.

Table 4 accounts for every planned selector row before retrieval is scored.

We compare nine ways to turn a selected ID into evidence. Two deterministic baselines use exact key lookup or exact decoded-title equality. The seven ranked rules include three BM25 variants over titles, bodies, or both, plus BGE-M3, ColBERTv2, reciprocal-rank fusion, and hybrid reranking (Robertson and Zaragoza, 2009; Chen et al., 2024; Santhanam et al., 2022; Beijing Academy of Artificial Intelligence, 2024). Exact

key lookup bypasses ranking and maps U directly to its passage. Exact title equality decodes the selected key and uses a verified one-to-one map from corpus titles back to object IDs. The ranked rules use that same decoded title as a query, then let every corpus object compete for the returned positions. These two anchors separate identity resolution from the ranking step.

Dense retrieval uses BGE-M3. ColBERTv2 compares query and passage tokens. Reciprocal-rank fusion combines both models with BM25 at $k = 60$ (Cormack et al., 2009). The final stack reranks the top 100 fused candidates. We request up to 20 returned IDs and score the recorded lists at cutoffs 1, 5, and 20.

All selector outputs and rankings were frozen before this audit. We score U and G against exactly the same returned IDs, so the target is the only part of the comparison that changes. Appendix B gives the selector prompt, execution settings, pinned model revisions, retriever settings, and full accounting.

3.2 The handoff rule determines object return

We hold the target fixed. On 1,463 records, the selected ID is resolvable and equals the dataset trace, so every rule is checked against the same object. All seven ranked rules receive the same one-to-one decoded selected title. This comparison isolates delivery of an explicit identity. Original-question retrieval is evaluated separately.

Table 5: Selected-object absence for two deterministic baselines and seven ranked retrieval rules. Every record has one fixed, resolvable target.

Handoff rule	Missing at 5	Missing rate
Exact key lookup	0/1,463	.000
Exact title equality	0/1,463	.000
Title-only BM25	66/1,463	.045
Title + body BM25	284/1,463	.194
Body-only BM25	389/1,463	.266
BGE-M3	65/1,463	.044
ColBERTv2	20/1,463	.014
Hybrid RRF	148/1,463	.101
Hybrid + reranker	14/1,463	.010

The deterministic rows verify that the resolver and title projection preserve identity on all 1,463 records. The seven ranked rules omit between 14 and 389 at cutoff five. Body-only BM25 loses the target on 26.6% of records, compared with 1.4% for ColBERTv2 and 1.0% for the final reranker. The selected ID and evaluation target stay fixed

Table 6: Cutoff-five misses by model family on the same aligned records used in Table 5. The target is fixed within every row.

Model	n	BM25	BGE	ColB.	RRF	Final
Claude	456	125	20	6	47	4
GPT	509	133	23	7	51	5
Gemini	498	131	22	7	50	5
All	1,463	389	65	20	148	14

Table 7: The 14 aligned records missed by the final reranker at cutoff five. Ranks are identical across the listed model rows. > 20 means absent from the released top 20.

Selected object	Rows	RRF	Final
Hypo-Arena	3	> 20	> 20
Philadelphia Athletics	3	5	7
Defender (association football)	3	8	8
Apple Vale, Queensland	3	> 20	> 20
SS Sylvan Arrow	2	> 20	> 20

throughout. The handoff rule determines whether that ID appears in the reader’s evidence list.

The pattern repeats across all three model families. Across the 1,463 aligned paired records, the final reranker retains 1,071 body-only BM25 hits and recovers 378 BM25 misses. It loses three BM25 hits, and both rules miss on 11 records. The comparison therefore isolates changes made by the evidence builder after selection.

The final reranker misses 14 selected objects across five questions. Six appear at rank seven or eight, beyond the reader’s cutoff. Eight are absent from both released top-20 lists. Under a documented selected-return requirement, six misses occur near the cutoff and are consistent with a cutoff adjustment. The remaining eight require deeper retrieval or explicit retention of the selected object to certify its return.

Appendix B.2 reports the corpus-wide title sanity check, remaining cutoffs, and per-model results.

3.3 Reading the final reranker misses

The final reranker leaves 14 misses in the aligned cohort. Every row in this analysis has a resolvable selected ID and $U = G$, so the selected-object and dataset-trace checks agree on the target. The misses come from five questions and form two clear groups in the released rankings.

Six misses sit just below the reader cutoff. Philadelphia Athletics appears at rank five after fusion and moves to rank seven after reranking for

Table 8: Target-aware cutoff-five BM25 results on the 329 records where U and G differ. Different is the sum of the two one-object columns.

Query	U		G		Different
	Both	only	only	Neither	
Original question	19	78	28	204	106
Direct rewrite	48	140	35	106	175
Retrieval fact	43	213	4	69	217
Decoded title	42	210	3	74	213

all three model records. Defender appears at rank eight before and after reranking. These objects remain in the recorded list, so changing the cutoff would change their handoff result.

The other eight misses are absent from both released top-20 lists. This group contains all three Hypo-Arena records, all three Apple Vale records, and two SS Sylvan Arrow records. The stored depth does not separate a rank below 20 from a complete lexical or neural nonmatch. ROP records the observable endpoint result without assigning an unseen cause.

3.4 Why the evaluation target matters

Across 1,792 resolvable model-question records, the selected and dataset-traced identities differ on 329 records drawn from 160 of the 600 questions. We score each frozen ranking twice, once against U and once against G . The ranking, cutoff, and identity resolver do not change. The same 600 questions appear once for each model family. Pooled model-question counts describe the recorded audit and are not independent question samples.

The four body-BM25 queries expose the selected ID at different points. The original question is shared across selector families, and U is not an input. The direct rewrite is generated from the question and full table, again without receiving U as the selected ID. The retrieval fact is co-generated with U and is prompted to name the same entity. Its executed BM25 terms match the decoded-title terms on 291 of the 329 object-distinct records. The decoded-title query is built directly from U using the fixed title projection. The last two forms are controlled selected-name delivery tests.

Original-question BM25 receives only the question, yet its two membership checks differ on 106 records, or 5.9% of the resolvable cohort. It returns neither object on 204 of the 329 object-distinct records. A direct rewrite gives 175 different labels. The retrieval-fact and decoded-title conditions deliberately name U and give 217 and 213. Their

larger counts show how query construction changes the magnitude and balance of target disagreement.

The selected-name stress test compares five stacks under the same decoded-title query. Selected-object return at cutoff five ranges from .725 to .992 across model families and stacks. Under the two targets, complete-cohort stack order remains unchanged for every model family and cutoff in the reported sensitivity checks. The target changes record-level diagnosis in this study, while aggregate stack choice stays stable. Appendix B.2 reports the full cutoffs, per-model breakdowns, and an exact ranking example.

4 From Returned Evidence to Answers

An object that reaches the evidence list can now influence the reader. We study this step on records where the selected and dataset-traced IDs agree, leaving one shared passage to follow from retrieval into the answer.

4.1 Passage presence and answer accuracy

The frozen answer matrix has three reader prompts with no answer hint. Their evidence comes from retrieval over the decoded selected ID, a model-written rewrite, or the original question. For each pair, we keep records where $U = G$, both prompts receive five passages, and the shared object appears in one passage list but not the other. Exact match uses standard HybridQA normalization.

Across all three pairs, exact match is 28.6 to 31.0 points higher in the context that contains the shared object. The comparison remains associative because the paired contexts also change passage identities, ranks, lengths, distractors, and input length. We therefore isolate one passage in a deletion experiment.

4.2 Deleting the aligned passage

The deletion study uses 64 $U = G$ items selected from earlier outputs. Each historical item was exact with the shared passage and inexact without it in both the retrieved-context and passage-only comparisons. We fixed these cases before the fresh calls, creating a high-contrast cohort for testing whether the same passage-specific pattern repeats.

Each item receives three intact repetitions and three trace-drop repetitions. The intact arm keeps the recorded request content. Trace-drop removes the aligned passage and leaves all other prompt content in place. For 59 items, sham-drop instead

Table 9: Hint-free answer associations when $U = G$. Return EM scores the context that contains the shared object, and Omit EM scores the paired context that does not. Intervals resample question IDs and keep all model-family rows for each sampled question together.

Arm pair	n	Return EM	Omit EM	Δ EM [95%]
ID/question	832	79.2	50.6	28.6 [24.5,32.8]
ID/rewrite	200	76.0	45.0	31.0 [23.5,38.9]
Rewrite/question	903	81.9	52.6	29.3 [25.4,33.4]

Table 10: Passage-deletion contrasts within the preselected 64-item high-contrast cohort. I, T, and S denote intact, trace-drop, and sham-drop. Values are three-repeat item means with paired bootstrap intervals.

Contrast	n	EM A/B	Δ EM [95%]
I/T	64	92.7/15.1	77.6 [68.8,85.4]
S/T	59	96.0/14.7	81.4 [71.8,89.8]
I/S	59	92.1/96.0	-4.0 [-9.0,1.1]

removes the nontraced passage closest in rendered length that does not contain the normalized reference answer. We measure the change in exact match between intact and trace-drop. Appendix B gives the task, retry, and failure accounting.

Within this fixed cohort, removing the aligned passage lowers exact match by 77.6 points, and sham-drop exceeds trace-drop by 81.4 points. The intact versus sham interval includes zero. All 24 terminal failures remain in their planned cells and score zero. Even under the most adverse reassignment of those failures, both trace-removal gaps remain at least 73.4 points. Majority exact is 63 of 64 for intact, 57 of 59 for sham-drop, and 9 of 64 for trace-drop. Answer-token F1 gives corresponding gaps of 48.5 and 52.1 points.

5 Related Work

Grounding research studies which evidence a language model retrieves and how that evidence supports its output. RAG supplies retrieved context for generation (Lewis et al., 2020). RAGChecker separates retrieval and generation errors, and MIRAGE links answer tokens to retrieved documents (Ru et al., 2024; Qi et al., 2024). CF-RAG uses counterfactual queries to distinguish causal evidence from correlated distractors, while AgenticRAGTracer diagnoses multi-step retrieval reasoning (Qin et al., 2026; You et al., 2026). Rationale studies test whether explanations reflect model behavior, while causal mediation measures how components affect a chosen outcome (DeYoung et al., 2020; Madsen

et al., 2024; Vig et al., 2020). These methods inspect evidence support and answer behavior. Our audit follows a named object across one component boundary before the reader sees it.

Retrieval inputs also change across a pipeline. ReAct, IRCoT, STEP, and Iter-RetGen use intermediate reasoning or generated text to guide later search (Yao et al., 2023; Trivedi et al., 2023; Luo and Surdeanu, 2022; Shao et al., 2023). Query rewriting, HyDE, and RaFe learn or generate new retrieval inputs (Ma et al., 2023; Gao et al., 2023; Mao et al., 2024). Our four query forms show that query construction must be reported alongside the profile because it determines how directly retrieval is exposed to the selected identity.

Identity-bearing grounding systems expose this handoff because they pass named entities or passages across stages. Entity linking resolves mentions to entity identities (Wu et al., 2020; Li et al., 2020). Graph and entity retrieval then use entity or passage identities to collect evidence for later components (Min et al., 2019; Sun et al., 2019; Shavarani and Sarkar, 2025). A unified data-to-text interface converts structured knowledge into retrievable text (Ma et al., 2022). HybridQA supplies tables with linked passages, and multi-row table-text QA draws evidence from both modalities (Chen et al., 2020; Kumar et al., 2023). OTT-QA retrieves tables and text, while TableRAG retrieves heterogeneous text and table chunks (Chen et al., 2021; Yu et al., 2025). ROP applies at a boundary only when the interface exposes one of these identities and documents that it must remain in the returned set.

Provenance, tracing, and software contracts provide general ways to record and validate pipeline state. PROV represents provenance information (World Wide Web Consortium, 2013). OpenTelemetry’s generative-AI conventions define retrieval fields (OpenTelemetry, 2026). Design by Contract and OCL express assertions and behavioral constraints (Meyer, 1992; Object Management Group, 2014). Typed data-validation systems encode type, presence, value-count, and domain constraints in schemas (Breck et al., 2019). ROP specializes these general mechanisms for a retrieval boundary by declaring the selected target, identity resolver, returned-ID field, cutoff, expected population, and verdict order. The reference evaluator scores every expected record and the complete run.

6 Discussion

The study begins with a simple interface fact. A reader can only use the passages it receives. Once an upstream component emits an object ID, the pipeline exposes a concrete boundary: U enters retrieval, and ranked IDs leave it. Table 5 shows that the rule between those two endpoints matters even when the query states the selected identity directly. Selected-object return measures this boundary and identifies where a recorded choice disappears.

The aligned and object-distinct analyses play different roles. On the 1,463 aligned records, U and G identify the same passage. The omission counts in Table 5 therefore measure the handoff rule without any target disagreement. This is the main implementation result. The 329 object-distinct records answer a reporting question: two valid membership tests can describe different components of the same ranking. The reader analysis returns to aligned records and follows one shared passage into the answer. These three pieces form one chain from selected ID to returned evidence to reader behavior.

A complete pipeline evaluation can place a score at each point in that chain. Selector evaluation records whether U fits the task. A selected-return miss identifies an endpoint omission after selection. A wrong answer after both checks pass leaves evidence quality and reader behavior to inspect. This division gives each score one component and one repair target.

Dataset-trace recall answers a second useful question. It asks whether retrieval covers the object linked by the benchmark. Selected-object return asks whether retrieval delivers the upstream choice. The 106 different labels under the original-question query show that these questions remain distinct even when retrieval never sees U . Reporting the target makes clear whether a score describes benchmark coverage or selector-to-retriever delivery.

Direct lookup and ranked retrieval serve different purposes. Lookup preserves the selected identity, while ranking adds neighboring passages or chunks. A pipeline with a documented selected-return requirement can reserve one reader slot for U and fill the remaining slots from ranked retrieval. ROP checks the reserved identity. Relevance and answer quality remain separate evaluations. The exact-lookup rows establish identity resolution, while the ranked rows measure what evidence construction retains or displaces.

Query construction must be reported alongside

the profile because it determines how directly retrieval is exposed to the selected identity. The decoded-title test exposes U directly, while the original question and direct rewrite do not receive it. The target disagreement changes from 106 records with the original question to 213 with the decoded title. That difference is part of the pipeline behavior. A reported handoff result therefore needs the target, query construction, returned-ID field, and cutoff together.

Complete population accounting closes a second gap. A pass fraction over rows that survived processing cannot reveal a missing row. The two-row example in Table 2 keeps the surviving pass fraction at 1.0, while ROP marks the incomplete run as unverifiable. This matters whenever API failures, parser errors, or interrupted jobs can remove planned records before aggregation.

ROP hard mode is tied to a documented selected-return requirement. For other interfaces, the same fields provide a descriptive audit without assigning a hard pass or failure. The synthetic fixture tests the evaluator and all verdict paths. The HybridQA study supplies the separate empirical stress test. Together, they show how to state the requirement, count the full run, and measure whether the selected identity reaches the reader.

7 Conclusion

An identity-bearing pipeline can hold a resolvable object ID and still lose it when building evidence for the reader. On 1,463 HybridQA records where the selected object matches the dataset trace, the two deterministic handoff rules return every object. The seven ranked rules omit between 14 and 389 at cutoff five. Measuring this boundary shows whether the recorded choice survives the rule that turns it into passages.

Dataset-trace recall measures benchmark coverage. Selected-object return measures whether the recorded choice survives retrieval. On identical original-question BM25 rankings, the two checks give different top-five labels on 106 of 1,792 resolvable records, or 5.9%. Query construction changes that disagreement rate. Together, the two targets identify which component the score describes.

ROP turns a documented selected-return requirement into a complete-run test. The reference evaluator binds the target, returned field, cutoff, resolver, and expected population, assigns a verdict to every record, and assigns a status to the complete run.

The HybridQA audit identifies which handoff rules omit the selected object. ROP makes the chosen rule and denominator explicit and replayable.

Limitations

ROP v1 checks endpoint membership in supplied, hash-bound records. It verifies byte consistency and agreement between observed IDs and the declared expected population. It does not establish when the declaration was created or who produced the logs. Those checks require external controls. The released hard-mode example uses a synthetic interface.

The empirical study covers one English Wikipedia table-text corpus and uses one recorded selector outcome per model-question pair, so it does not estimate selector stability. The 329 exact-ID mismatches have no semantic annotations. The deletion result applies to its deliberately selected 64-item cohort. Selector quality, whether both identities provide valid evidence, and average answer benefit require separate evaluations.

Ethical Considerations

The study uses the public, Wikipedia-derived HybridQA dataset and collects no new personal data. We recruited no human annotators. We do not treat the generated answers as verified facts or evaluate them for social bias. The release retains invalid outputs and item-level traces for inspection.

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A Audit Records and Replay

A.1 How ROP Evaluates a Run

ROP evaluates one handoff. An upstream component supplies a selected key or declared alternatives, and a downstream component supplies ranked raw corpus IDs. Each run begins with a declaration, the instruction sheet that fixes what the evaluator will check. In hard mode, the declaration also lists every record ID expected in the run. The evaluator resolves the identities, tests endpoint membership, assigns one verdict to each observed record, accounts for missing expected IDs, and assigns one run status.

Set the mode and population: The declaration names the target interface, interface version, policy authority, producer, and policy revision. Hard mode applies exactly when the documented interface specification requires the selected object to remain available at the downstream endpoint. A valid soft or open declaration returns NOT_APPLICABLE. Any accompanying measurements remain separate diagnostics. Prompt text preserves the prompt used for the run. The declaration alone determines the mode.

ROP v1 uses the authority and producer named in the declaration. SHA-256 hashes verify that the evaluated files contain the same bytes as the supplied files. They do not authenticate the producer. Independent verification of event order and timing requires signed timestamps. Signed timestamps are not part of v1.

In hard mode, the expected population is the complete list of record IDs required for the run. The declaration stores this population as a sorted list, its count, identifier field, and SHA-256 hash. Its metadata states that the list was fixed before returned IDs were produced. ROP verifies the supplied bytes and internal consistency. Timing is established outside the evaluator. The evaluator compares every observed row with that list and retains every row. Missing IDs appear in `missing_record_ids` and in the MISSING histogram. The evaluator also detects duplicate IDs, malformed IDs, and unexpected records. A diagnostic pass fraction uses the evaluable records. The run status uses the declared population and accounts for every expected ID.

The JSON Schema validates manifest syntax. The evaluator then validates relationships across files and fields. The declaration envelope is the top-level wrapper around the instruction sheet. These checks cover the envelope, every SHA-256 hash, interface and component equality, mode and policy revision, population order and count, resolver ownership, whether each witness decodes to a distinct selection, and every referenced file. The schema and these semantic checks together determine the evaluation result.

Resolve identities and test the ranking: A versioned resolver D_v maps each raw key, alias, or chunk ID to one canonical object. Let $\mathcal{U}_i$ contain the selected key or its declared alternatives. The evaluator reads the first k raw IDs returned by the component named in the declaration, where $\hat{R}_i@k = (r_1, \dots, r_k)$. Every value in $\mathcal{U}_i$ must resolve, giving $S_i = \{D_v(u) : u \in \mathcal{U}_i\}$. ROP applies the following rules to each canonical object o :

$$q_{\text{any}}(o, \hat{R}@k) = \mathbf{1}[\exists r \in \hat{R}@k : D_v(r) = o], \quad (1)$$

$$q_{\text{all}}(o, \hat{R}@k) = \mathbf{1}[M_v(o) \subseteq \text{set}(\hat{R}@k)]. \quad (2)$$

Here $M_v(o)$ is the resolver’s nonempty set of raw members for object o . The evaluator applies the declared rule to one acceptable object at a time. The membership test succeeds exactly when $q(o, \hat{R}@k) = 1$ for at least one $o \in S_i$. Under any,

aliases, object IDs, and member IDs first map to their canonical object, and one match is sufficient. Under all, every raw resolver member for one acceptable object must appear. The evaluator applies the cutoff before either rule. An alias or member assigned to two canonical owners invalidates the resolver. Unknown returned IDs are nonmatches. Duplicate IDs keep their ranked positions and consume cutoff slots.

In v1, the string fields `corpus_uri` and `composite_corpus_key` are passed to the resolver without interpretation. The evaluator only applies their declared nonempty-string constraint.

Decode a bound witness: An evaluation record carries identity in one of two forms. It either stores the raw selected key or stores a declared invertible witness with a hash-bound exact inverse map. A bound lookup table decodes the witness back to the selected key. In witness mode, each record carries a SHA-256 hash over the binding profile, declaration ID and exact manifest hash, population file hash and record ID, witness value, decoded selection, and inverse-map file hash. The lookup table must be one-to-one: no two witness values decode to the same selection. An invalid or non-injective lookup table, or a decoded selection outside the resolver, produces `DECLARATION_INVALID`. An unknown witness value produces `RESOLUTION_ERROR`. A missing, malformed, or incorrectly hash-bound record witness produces `SELECTOR_ERROR`. These rules establish endpoint membership from either the carried selected key or the selection decoded from the bound witness. The hashes verify the supplied bytes and do not authenticate the producer. A lineage extension that records each intermediate hand-off and its verified parent remains future work. The endpoint check returns either a membership label or a specific identity, resolution, or returned-log error. The complete ROP evaluator validates the declaration and files, accounts for every expected ID, assigns record verdicts, and produces the run status.

Assign one verdict to each record: The evaluator applies the ordered checks in Table 11. Some records receive a terminal verdict before the membership test. Each observed record receives exactly one of seven terminal verdicts: `PASS`, `FAIL`, `SELECTOR_ERROR`, `RESOLUTION_ERROR`, `UNVERIFIABLE`, `DECLARATION_INVALID`, or `NOT_APPLICABLE`. The first matching condition

Record verdict	First condition that applies
<code>DECLARATION_INVALID</code>	Top-level declaration envelope, instruction sheet, authority, policy, population, resolver, identity witness, output binding, or any referenced file fails validation
<code>NOT_APPLICABLE</code>	Valid declaration selects soft or open mode
<code>SELECTOR_ERROR</code>	Selected key or bound witness is missing, malformed, or incorrectly bound
<code>RESOLUTION_ERROR</code>	Selected identity is absent from the verified resolver
<code>UNVERIFIABLE</code>	Returned-ID field is missing or malformed
<code>PASS</code>	Declared endpoint-membership rule is satisfied
<code>FAIL</code>	Declared endpoint-membership rule is violated

Table 11: Record verdicts in fixed order. The first condition that applies assigns the final verdict.

assigns the final verdict, and no later check overwrites it. In hard mode, every expected ID is represented by an observed record or by an explicit `MISSING` entry in the run histogram and missing-ID list.

Assign one status to the run: After assigning record verdicts, the evaluator gives the complete run one of five statuses. It first validates the declaration. A valid soft or open declaration yields `NOT_APPLICABLE`. In hard mode, the evaluator compares the observed IDs with the complete expected list and then combines the expected record verdicts. Incomplete evidence receives its own run status.

Requirements for `RUN_PASS`: For a hard run, `RUN_PASS` requires every check to succeed. Every referenced file must match its SHA-256 hash. The declaration, instruction sheet, policy, population, resolver, and any witness must pass their semantic checks. The observed IDs must exactly equal the declared expected population. Each expected ID appears once, with no malformed, duplicate, or unexpected ID. Every expected record must satisfy the declared return condition. The run status remains separate from the diagnostic pass fraction.

Fields retained for replay: The released ledgers join on question identifier and, where applicable, model family. Each bound row retains the selected identifier U_i , dataset-traced identifier G_i , and raw

Run status	First condition that applies
RUN_INVALID	Declaration invalid, or a hard-mode run contains malformed, duplicate, or unexpected observed IDs
NOT_APPLICABLE	Valid declaration selects soft or open mode
RUN_FAIL	Hard mode, no invalid-run condition, and at least one expected row has verdict FAIL
RUN_UNVERIFIABLE	Hard mode, no invalid-run condition, and no FAIL. An expected ID is missing or an expected row has selector error, resolution error, or a missing or malformed returned-ID list
RUN_PASS	Hard mode, observed population equals the sealed population, and every expected record passes

Table 12: Run statuses follow a fixed order. An invalid declaration yields RUN_INVALID. A valid soft or open declaration yields NOT_APPLICABLE. A hard run is then classified by population validity, recorded failure, incomplete evidence, or complete success.

retrieved document IDs $\hat{R}$ for every query and retriever. The stored ranking depth covers every reported cutoff. The versioned resolver records corpus-resolution status and maps the raw IDs to the object ranking R . Replay derives resolved identity agreement C , selected-object output membership P , and trace-object output membership H . The API-call ledgers retain terminal-attempt status and explicit invalid-output fields. The artifact manifest and replay extractors bind each exact source member by byte count and SHA-256 hash. The answer ledgers retain the ordered evidence links and normalized answer scores used by the matched-five audit.

Validate joined records before analysis: Before a joined row enters an analysis, the replay tool validates its bound source rows and reconstructs every field used by that analysis. All rows must share the same question and, where applicable, the same model. Replay also checks the named condition, reference path, cutoff, logical task, terminal attempt, invalid-output policy, source SHA-256 hashes, and status. A mismatch rejects the joined row and leaves the declared denominator unchanged. A schema-invalid terminal response or unissued dependency cell remains explicit under the recorded denominator policy. The same released bytes and deterministic extractors reproduce the joined rows and aggregate counts.

Measure overlap from joined rows: Separate

field totals do not identify their overlap. The released rows compute every reported conjunction directly on a shared question-model key. The analysis counts complete joined records. This study binds one dataset-traced path per item. The executable selected-return profile declares set-valued selection rules before scoring and accepts alias- or chunk-level returned IDs. An evaluation with several acceptable benchmark traces must declare the trace set before scoring and retain every within-path conjunction.

B Reproducibility Details

This appendix records cohort construction, retrieval outputs, model settings, and the open-weight check.

B.1 Cohort construction

We build the cohort from fixed rules that do not use model outputs. We normalize each question by lowercasing its text, keeping matches of $[a-z0-9]^+$, and joining them with one space. We apply this case-insensitive term list: *most, least, highest, lowest, largest, smallest, biggest, oldest, youngest, first, second, third, last, latest, earliest, more than, less than, finished, winner, won, gold, silver, bronze, position, and ranked*. A salted SHA-256 hash of the question identifier orders every item that passes the filters. We take the first 600, allowing at most one question from any table.

The released files make this selection reproducible. They record the salt, the source hash, each item’s selection hash, and the final cohort hash. They also freeze the IDs and hashes for all 7,201 training candidates and 455 development candidates. The split denominators come directly from the pinned upstream HybridQA files.

Keeping protocol development separate: We reserved a hash-locked set of 120 questions for protocol development and excluded those IDs from every later development cohort. These later cohorts provide a separate in-dataset check. Before collecting the corresponding model outcomes, we fixed the cohorts, prompts, answer conditions, retrieval cutoffs, scoring rules, and treatment of invalid responses. The prompt also fixed what `selected_link` means.

Frozen selector instruction: The system instruction states:

You select a bridge entity from a table. The bridge entity is the one table-linked Wikipedia

page whose passage is needed to answer the question. Do not answer the question. Use the table and the question to choose exactly one link that is explicitly shown. Return only the required JSON object.

The question and full table appear before this final user instruction:

Choose the single shown link that best bridges the table to the external information needed for the answer. Set `selected_link` to the exact `/wiki/...` value shown. Set `retrieval_fact` to a concise human-readable name for that same entity, suitable as a search query. Do not include the final answer.

This instruction defines U for the retrospective comparison.

Synthetic hard-mode fixture: The released hard fixture uses `fixture-ranker-api/v1`. It reads the selected target from `selector.selected_key`, reads returned IDs from `object_id`, checks the first three under any, and expects exactly `run_alpha` and `run_beta`. The prompt is stored as provenance and cannot activate or override hard mode.

Every analysis that compares the selected key with the dataset trace reuses the same frozen identifiers, frozen ranking records, and answer records. The Qwen analysis and the checks that use object identifiers directly have their own hash-bound inputs. For each analysis, the release connects the exact inputs to the code, execution metadata, and outputs that produced the reported result.

B.2 Retrieval results across models and cutoffs

The decoded-title query is a controlled selected-name test. It exposes U through the verified one-to-one title projection and asks whether each stack returns that identity.

Across the complete corpus inventory, exact title equality returns every object at all three cutoffs. Title-only BM25 omits 15,379 objects at cutoff one, 7,667 at cutoff five, and 5,258 at cutoff 20. Of the 286,270 title queries, 141 become empty after lexical preprocessing. This is a corpus-wide identity-resolution sanity check without a selector.

One frozen record shows how the target check works. The selector chooses Xavier University as U , while HybridQA links Xavier Musketeers football as G . With the shared query Xavier University, the two frozen handoff rules in Table 15 give opposite top-five labels.

Table 13: Corpus-wide omission counts when every object is queried by its own decoded title. Each column has denominator 286,270.

Handoff rule	At 1	At 5	At 20
Exact title equality	0	0	0
Title-only BM25	15,379	7,667	5,258

Table 14: Selected-object return in the decoded-title test at three cutoffs. Denominators are 599 for Claude, 594 for GPT, and 599 for Gemini.

Retriever	Model	R@1	R@5	R@20
Body BM25	Claude	.506	.725	.840
	GPT	.515	.746	.845
	Gemini	.524	.750	.855
BGE-M3	Claude	.778	.955	.983
	GPT	.773	.958	.988
	Gemini	.775	.955	.988
ColBERTv2	Claude	.881	.983	.990
	GPT	.896	.988	.992
	Gemini	.891	.987	.992
Hybrid RRF	Claude	.674	.891	.967
	GPT	.695	.897	.966
	Gemini	.699	.902	.972
RRF + rerank	Claude	.855	.990	.995
	GPT	.855	.992	.995
	Gemini	.856	.990	.995

Handoff rule	Rank of U	Rank of G	Top-five result
Body-only BM25	6	3	G only
BGE-M3	1	16	U only

Table 15: Frozen rankings for one Claude record. U is Xavier University and G is Xavier Musketeers football.

Table 16: Cutoff-five results for the 329 object-distinct records under the decoded-title query. Different labels is the sum of the two one-object columns.

Stack	U		G		Different labels
	Both	only	only	Neither	
Body BM25	42	210	3	74	213
BGE-M3	31	284	0	14	284
ColBERTv2	39	285	0	5	285
Hybrid RRF	34	258	0	37	258
RRF + rerank	48	278	0	3	278

Stack	k	Claude	GPT	Gemini	Pooled
Body-only BM25	1	0/83/2/58	0/54/2/29	0/67/1/33	0/204/5/120
	5	15/88/2/38	13/54/0/18	14/68/1/18	42/210/3/74
	20	22/95/0/26	20/50/0/15	19/69/0/13	61/214/0/54
BGE-M3	1	0/115/0/28	0/72/0/13	0/82/0/19	0/269/0/60
	5	11/125/0/7	10/73/0/2	10/86/0/5	31/284/0/14
	20	16/122/0/5	16/68/0/1	14/86/0/1	46/276/0/7
ColBERTv2	1	0/127/0/16	0/82/0/3	0/94/0/7	0/303/0/26
	5	13/126/0/4	13/72/0/0	13/87/0/1	39/285/0/5
	20	25/116/0/2	21/64/0/0	20/81/0/0	66/261/0/2
Hybrid RRF	1	0/97/1/45	0/67/0/18	0/80/0/21	0/244/1/84
	5	12/113/0/18	11/64/0/10	11/81/0/9	34/258/0/37
	20	20/116/0/7	20/61/0/4	18/81/0/2	58/258/0/13
RRF + rerank	1	0/129/0/14	0/80/0/5	0/94/0/7	0/303/0/26
	5	14/127/0/2	17/68/0/0	17/83/0/1	48/278/0/3
	20	26/116/0/1	24/61/0/0	23/78/0/0	73/255/0/1

Table 17: Complete object-distinct counts across every frozen search stack and cutoff. Each cell lists both objects returned, selected object only, trace object only, and neither object, in that order: $P_1 H_1 / P_1 H_0 / P_0 H_1 / P_0 H_0$. Model denominators are 143, 85, and 101. The pooled denominator is 329 model-question records from 160 questions. Pooled rows are descriptive because records sharing a question are dependent.

Across all 1,792 resolvable selections, the final reranker returns U on 1,775 records and G on 1,497. The labels differ on 278. Within the 329 object-distinct records, moving from body-only BM25 to the final reranker changes 74 selected-object labels from miss to hit and none from hit to miss. The selected-object return rate rises by 22.5 points, with a 95% question-cluster bootstrap interval of [16.3, 29.3] points. This is a selected-name stress test.

The remaining tables expand the joint outcomes, omission mechanisms, and complete cutoff grid.

Stack	$k = 1$	$k = 5$	$k = 20$
Body BM25	209	213	214
BGE-M3	269	284	276
ColBERTv2	303	285	261
Hybrid RRF	245	258	258
RRF + rerank	303	278	255

Table 18: Number of object-distinct records receiving different selected-object and trace-object labels. Every cell has denominator 329.

Results across every cutoff: The full retrieval matrices show that query choice changes the returned IDs beyond cutoff five. With BM25, a question query recalls the dataset trace at .087/.278 for cutoffs 1/20. Queries built from the decoded selected ID give .370 to .423 at cutoff 1 and .680 to .753 at cutoff 20. At cutoff five, question recall is .183. The corresponding Claude/GPT/Gemini values are .580/.648/.637 for decoded selected-ID queries, .590/.647/.637 for model-written facts, and

Model	Absent at 5	Ranks 6 to 20	Nonmatch	Unresolved
Claude	165	69	5	91
GPT	151	59	5	87
Gemini	150	63	4	83

Table 19: Partition of body-only BM25 top-five non-membership. Absent at 5 is the total, and the other three columns partition it. Nonmatch is verified when fewer than 20 documents satisfy the OR query and the selected passage is absent. A full 20-result list cannot separate a lower rank from nonmatch.

Retriever	Q	Claude	GPT	Gemini
BM25	.183	.580	.648	.637
BGE-M3	.183	.745	.827	.812
ColBERTv2	.185	.772	.858	.840
Hybrid RRF	.230	.702	.782	.767
RRF + rerank	.358	.777	.868	.852

Table 20: Traced-passage $R@5$ across retrieval stacks. Every rate has denominator 600. Q uses the 600 unique original questions and is shared across models. Each model column queries with that model’s selected decoded-ID query, with invalid upstream rows counted as misses.

.667/.740/.645 for direct rewrites. Across BGE-M3, ColBERTv2, hybrid RRF, and hybrid reranking, each model’s decoded-ID query also exceeds the question at both cutoffs. The replay retains the result for every model, retriever, and cutoff.

The matched-five answer audit behind Table 9 uses complete terminal rows from the frozen answer matrices. A record enters the comparison only when its experimental arm, question and model pairing, five-passage context, no-hint prompt, selected-passage return label, normalized exact-match score, denominator, and question-cluster assignment match the frozen protocol.

B.3 Trace-deletion call settings

The trace-deletion experiment calls google/gemini-3.1-pro-preview through an API gateway pinned to Google AI Studio with provider fallback disabled. The settings are temperature 1.0, medium reasoning, strict one-field JSON, no seed, three repeats, and at most two attempts. Three successful preflight calls tested the route and response schema and are excluded from analysis. The 561 main call tasks were released together for parallel execution, and retries produced 570 provider requests; 24 tasks ended in failure. The intact arm sends the same message content as the bound historical requests, using a different transport and schema serialization. The failures comprise 23 provider error envelopes and one invalid JSON response. They are distributed as 13 intact, 8 trace-drop, and 3 sham-drop tasks. The intact and sham F1 ordering changes under complete-case and valid-repeat summaries. Both trace-removal effects remain.

Table 21: Per-arm call accounting and repeat stability. Provider requests include retries. Same answer means three valid identical normalized outputs, and majority exact means an exact answer in at least two repetitions.

	Intact	Trace-drop	Sham-drop
Items	64	64	59
Scheduled calls	192	192	177
Provider requests	194	194	182
Valid / failed calls	179 / 13	184 / 8	174 / 3
Same valid answer, 3/3	51/64	43/64	54/59
Majority exact	63/64	9/64	57/59

The ranking manifest pins BAAI/bge-m3 at revision 5617a9f6, colbert-ir/colbertv2.0 at revision c1e84128, and BAAI/bge-reranker-v2-m3 at revision 953dc6f6. The corresponding model families are described in (Chen et al., 2024; Santhanam et al., 2022; Beijing

Academy of Artificial Intelligence, 2024).

Larger development retrieval check: A separate 1,397-item cohort tests text projection across the same five stacks. It contains 1,182 records without a model selection and 215 whose answer appears in the table. Starting from the dataset trace, we compare original-question and decoded-ID retrieval.

Retriever	$R@5(Q)$	$R@5(ID)$	Difference
BM25	.213	.765	.552
BGE-M3	.192	.957	.765
ColBERTv2	.236	.992	.756
Hybrid RRF	.243	.892	.649
Hybrid + rerank	.359	.993	.634

Table 22: Cutoff-five trace return on the 1,397-item development cohort. Q uses the original question, and ID uses the decoded dataset-traced identifier.

B.4 Open-weight Qwen check

The open-weight check asks whether the selected key survives the same retrieval handoff with Qwen3-32B-AWQ upstream. We use $C = 1$ when the selected object and dataset trace resolve to the same object, and $C = 0$ otherwise. Revision 0499c3ac83fdef88 uses reasoning, a 4,096-token cap, temperature .6, top- p .95, top- k 20, presence penalty 1.5, and global seed 12,345 with deterministic per-task derivation. One parser-valid selected link falls outside the displayed candidates and remains explicit.

Subset	Resolvable	$k = 1$	$k = 5$	$k = 20$
Final hybrid, all	334	294	330	332
Final hybrid, $C = 1$	229	204	226	228
Final hybrid, $C = 0$	105	90	104	104
Body BM25, all	334	165	231	282
Body BM25, $C = 1$	229	113	160	195
Body BM25, $C = 0$	105	52	71	87
Exact decoded field	334	334	334	334

Table 23: Selected-object return in the open-weight check. Final hybrid uses Qwen’s recorded retrieval fact. Body BM25 uses the decoded selected ID and reuses a hash-bound ranking only for an identical FTS5 query. The cutoff columns report returned-object counts.

Joint outcomes at cutoff five: For all 334 resolvable records, $(P_1 H_1, P_1 H_0, P_0 H_1, P_0 H_0) = (232, 98, 0, 4)$. On the 105 $C = 0$ records, the same table is $(6, 98, 0, 1)$. One additional $C = 0$ key is unresolvable and has $H = 0$. Released hashes bind the rows, tasks, rankings, and resolver.